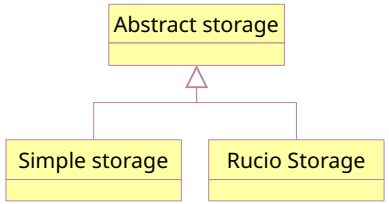
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RFC 9457 problem details

ExecutionBroker error responses
Evaluation and prototype

Dave Morris
Manchester University



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generated by @deepseek-ai/dsh (deepseek-flash)

Dave Morris
dave.morris@manchester.ac.uk

Background

Errors were told apart by HTTP status alone.

- Invalid update: logged, session unchanged, HTTP 200.
- Unknown session or offer set: an empty 404.
- Foreign session update: an empty 403.
- A client could not tell an invalid update from success.

Seven TODOs in the broker pointed at issue #90.

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Dave Morris
dave.morris@manchester.ac.uk

RFC 9457

One JSON object describes any HTTP error.

```
type      URI reference; defaults to about:blank
title     short summary of the problem type
status    HTTP status code for this occurrence
detail    explanation for this occurrence
instance  URI reference for this occurrence
```

The media type is `application/problem+json`.

Extension members are allowed.

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Gap

Problem details give the client something to branch on.

- A stable type URI instead of prose to parse.
- Title and detail for the human, instance for the request.
- Extension members carry IVOA diagnostics.
- The error model lives in the same OpenAPI document.

Generated Java and Python clients get the model.

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Prototype

The schema and the broker now cover every error path.

- ProblemDetails schema with a messages extension.
- Error responses documented on the affected paths.
- ProblemDetailsAdvice maps engine and web exceptions.
- Spring Security 401 and 403 share the same body.
- Schema and packages move from 1.0.8 to 1.0.9.

46 Java tests and the Python suites cover these paths.

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Mapping

One status code and one problem type per condition.

```
400  urn:ivoa:calycopis:error:invalid-update
400  urn:ivoa:calycopis:error:execution-rejected
401  urn:ivoa:calycopis:error:unauthorized
403  urn:ivoa:calycopis:error:access-denied
404  urn:ivoa:calycopis:error:session-not-found
404  urn:ivoa:calycopis:error:offerset-not-found
500  urn:ivoa:calycopis:error:internal-error
```

Framework errors keep their status and use `about:blank`.

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Example

An unknown session returns a 404 problem document.

```
GET /sessions/0000-...-0000
Accept: application/xml
404 Not Found
Content-Type: application/problem+json
{
  "type": "...:error:session-not-found",
  "title": "Execution session not found",
  "status": 404,
  "detail": "No session with UUID [...].",
  "instance": "/sessions/0000-...-0000"
}
```

The request asked for XML. The error is still JSON.

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dave.morris@manchester.ac.uk

Example

An invalid update returns 400 with structured messages.

```
400 Bad Request
Content-Type: application/problem+json
{
  "type": "...:error:invalid-update",
  "detail": "Unsupported path [/bogus].",
  "messages": [
    { "level": "ERROR",
      "values": { "path": "/bogus" } }
  ]
}
```

`messages` reuses the existing `MessageItem` model.

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Trade-offs

Two choices are left to the IVOA.

- Errors are JSON even when the client asks for XML.
- Honouring `Accept` needs a content selector and more schema.
- The test matrix roughly triples for three formats.
- A rejected direct execution stays a domain result.
- An opt-in flag switches it to a problem document.

A client that only accepts XML now gets JSON errors.

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dave.morris@manchester.ac.uk

Recommendation

Adopt RFC 9457 and settle the open questions.

- Adopt RFC 9457 and `application/problem+json`.
- Define a problem type namespace and registry.
- Decide the XML and YAML representation.
- Keep domain results separate from protocol errors.
- Recommend the `messages` extension for diagnostics.

Open questions: the type registry, XML and YAML errors, the `messages` member, and the domain result choice.

Issues: `uksrc/Calycopis-broker#90`
Discussion: `ivoa-std/ExecutionBroker`
Please join the discussion.

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Dave Morris
dave.morris@manchester.ac.uk

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Thank you

GitHub 

https://github.com/ivoa-std/ExecutionBroker

Please join the discussion.

We need your help to make it happen.


```
AIMetrics: [
  { "timestamp": "2026-10-20T15:07:00",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 100, "units": "%" } },
  { "interval": "2026-10-20T15:19:00/2026-10-20T15:20:00",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 5, "units": "%" } },
  { "interval": "2026-10-20T15:24:00/2026-10-20T15:32:00",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 3, "units": "%" } },
  { "interval": "2026-10-20T15:44:00/2026-10-20T15:49:00",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 2, "units": "%" } },
  { "interval": "2026-10-20T16:00:00/2026-10-20T16:02:00",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 2, "units": "%" } },
  { "interval": "2026-10-20T16:04:00/2026-10-20T16:05:56",
    "name": "@deepseek-ai/dsh", "version": "0.2.0-rc.2",
    "model": "deepseek-flash",
    "contribution": { "value": 6, "units": "%" } }
  ]
```


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dave.morris@manchester.ac.uk